IMPERIAL

Rotating Wave Approximation, Quantum Gates and the Quantum Fourier Transform

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Introduction

In this poster, we will:

- Justify the use of a Rotating Wave Approximation Hamiltonian
- Construct Quantum Logic Gates, starting with the Phase and Hadamard
- Look at an application of Quantum Logic Gates, the Quantum Fourier Transform

We want to introduce the Pauli matrices (Barnett 2026b, p. 15), Physical/Exact Hamiltonian and Rotating Wave Approximation (RWA) (Barnett 2026b, p. 18) to obtain quantum gates.

Pauli matrices:

$$\sigma_X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Physical Hamiltonian (Barnett 2026b, p. 15):

$$\mathcal{H}_{exact}(t) = \frac{\epsilon}{2}\sigma_Z + \gamma\sigma_X \cos(\omega t), \qquad \epsilon, \gamma, \omega \in \mathbb{R}_{\geq 0}$$

In order to obtain the RWA, "it is customary to work entirely within the frame rotating at ω". (Barnett 2026b, p. 18)
Rotating Frame:

$$|\psi'(t)\rangle = e^{i\frac{\omega t}{2}\sigma_Z}|\psi(t)\rangle, \qquad t \in \mathbb{R}_{\geq 0}$$

ROT/RWA:

$$\mathcal{H}_{rot}(t) = \frac{\epsilon - \omega}{2}\sigma_Z + \frac{\gamma}{2}\sigma_X + V(t), \qquad \mathcal{H}_{RWA} = \frac{\epsilon - \omega}{2}\sigma_Z + \frac{\gamma}{2}\sigma_X$$

$$V(t) = \frac{\gamma}{2}(\sigma_X cos(2\omega t) - \sigma_Y sin(2\omega t)$$

Is the RWA viable?

Is the $\mathcal{H}_{RWA}$ a close approximation to $\mathcal{H}_{exact}$ when we drop V(t)?

We must start with solving the Time-dependent Schrödinger Equation (Barnett 2026b, p. 9):

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \mathcal{H}(t)|\psi(t)\rangle$$

$\hbar$ = 1 for all calculations in this poster.
We use:

$$|\psi_{RWA}(t)\rangle = |\psi_{RWA}(0)\rangle e^{-i\mathcal{H}_{RWA}}, \qquad |\psi_{exact}(t)\rangle = \mathcal{T}exp(-i\int_0^t \mathcal{H}_{exact}(t')dt')|\psi_{exact}(0)\rangle$$

for an inner product, with $\mathcal{H}$ being picked as the Exact and RWA Hamiltonian for each function, $\mathcal{T}$ is the time-ordering operator, derived from a Dyson series (Cappellaro 2012, p. 4).

We define an inner product function on Python and compute the time-averaged state Fidelity (how close 2 Quantum operations are to each other (Ivezic 2023)):

$$\bar{F} = \frac{1}{T}\int_0^T F(t)dt, \qquad F(t) = |\langle\psi_{exact}(t)|\psi_{RWA}(t)\rangle|^2, \qquad 0 \leq \bar{F} \leq 1, T \in \mathbb{R}_{\geq 0}$$

Deriving F(t) from Born Rule (Barnett 2026a, p. 11), the $\bar{F}$ is calculated for $T = 10$ using the Trapezium rule, using 10,000 data points and a NumPy array (Max. Fidelity Error = 1 - min(Fidelity)).

Table 1: Fidelity and Max Fidelity Error (M.F.E). Obtained using NumPy (Harris et al. 2020) and SciPy (Virtanen et al. 2020) (trapezoid function).

ϵ	γ	ω	Fidelity	Max. Fidelity Error
11	1	10	0.9980863	0.0066193
41	1	40	0.9998697	0.0004489
1001	1	1000	0.9999995	1.245E-6

When $\sqrt{(\epsilon - \omega)^2 + \gamma^2} \ll \omega$ is obeyed (which is required for negligible oscillations (Barnett 2026b, p. 17)), the solutions to the Schrödinger Equation are extremely similar, so we can drop V(t).

Realised Single Qubit Gates

Using our Rotating Wave Approximation, we must obtain realised gates of the form $U(\tau) = e^{-i\mathcal{H}_{RWA}\tau}$ (Barnett 2026b, p. 20) and compare U to the Sought form of a quantum gate. What parameters do we substitute in order to obtain the correct quantum logic gates?

Table 2: Realised Gates

Gate	Sought Form	γ	τ	ϵ
$Phase$	$e^{-i\phi\sigma_Z}$	0	$\frac{2\phi}{\epsilon-\omega}$	$\ll \omega$
$Hadamard$	$\frac{1}{\sqrt{2}}(\sigma_X + \sigma_Z)$	$\epsilon - \omega$	$\frac{\pi}{\sqrt{2}\gamma}$	$\ll \omega$
X/NOT	σ_X	$\neq 0$	$\frac{(2k+1)\pi}{\gamma}$	ω
$\sqrt{X}$	$\sqrt{\sigma_X}$	$\neq 0$	$\frac{(2k+1)\pi}{2\gamma}$	ω
$\frac{\pi}{8}/T$	$e^{-i\sigma_Z\frac{\pi}{4}}$	0	$\frac{\pi}{2(\epsilon-\omega)}$	$\ll \omega$
S	$e^{-i\sigma_Z\frac{\pi}{2}}$	0	$\frac{\pi}{\epsilon-\omega}$	$\ll \omega$
$R_X(\theta)$	$e^{\frac{-i\sigma_X\theta}{2}}$	$\neq 0$	$\frac{\theta}{\gamma}$	ω
$R_Y(\theta)$	$e^{\frac{-i\sigma_Y\theta}{2}}$	$\neq 0$	$\frac{\theta}{\gamma}$	ω

These are correct up to global phase, which is physically irrelevant.
$R_X(\theta), R_Y(\theta)$ (derived via identity) are the Rotation gates with a respective axis X and Y. $\theta \in \mathbb{R}$
$U(\theta, \phi, \lambda)$ is the General Single Qubit Rotation Gate defined as $R_Z(\phi)R_Y(\theta)R_Z(\lambda)$ $(\phi, \theta, \lambda \in \mathbb{R})$. Can be derived using the ideas of rotations. (Nielsen and Chuang 2010, p. 174-177)
Picking $\phi = \frac{\theta}{2}$ in the Phase gate yields $R_Z(\theta)$.
$k \in \mathbb{Z}$ for the X/NOT and $\sqrt{X}$ gates.

It is also very important to know the Dyadic Rational Phase Gate (Nielsen and Chuang 2010, p. 218):

$$R_k = \begin{pmatrix} 1 & 0 \\ 0 & e^{\frac{2\pi i}{2^k}} \end{pmatrix}$$

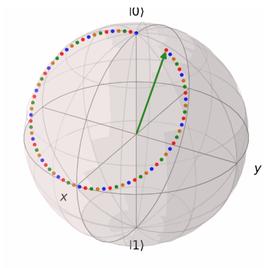
We later use this gate for Quantum Fourier Transforms, with $R_1 = Z, R_2 = S, R_3 = \frac{\pi}{8}/T$ gate. $(k \in \mathbb{N})$

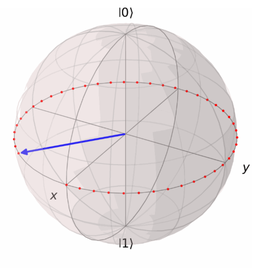
The Bloch Sphere

It is helpful to be able to visualise these gates, and a useful way of doing this is the Bloch Sphere. To get the idea:

- σ_X, σ_Y and σ_Z are π radians anticlockwise rotations, up to a global phase, in their respective axis.
- The Phase gate is an anticlockwise rotation about the Z axis, angle ϕ.

Below, the Hadamard gate has been applied twice on $|0\rangle$ and the Phase gate has been applied on $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. It can be seen below that $H^2 = I$.





These are produced as GIFs using Python's QuTip (Lambert et al. 2026), NumPy (Harris et al. 2020), ImageIO (ImageIO Contributors 2026) and OS packages.

Quantum Fourier Transform

Now we want to introduce the 2-qubit SWAP gate by name, which will be important in the application of the Quantum Fourier Transform (QFT).

We define the QFT by using the Discrete Fourier Transform. The QFT (Nielsen and Chuang 2010, p. 217) definition is seen below:

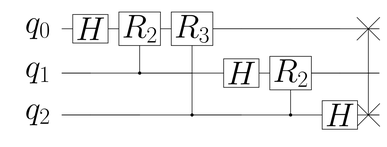
$$|j\rangle \equiv \frac{1}{\sqrt{N}}\sum_{k=0}^{N-1} e^{\frac{2\pi ijk}{N}}|k\rangle$$

Where N: Length of state, $|k\rangle$: a complex state of input data, $|j\rangle$: a complex state of transformed data.
When computing a Quantum Fourier transform, we want to take $N = 2^n$ where n is the number of qubits in a Quantum Computer.

Writing $|j\rangle = |j_1, ..., j_n\rangle$ in a binary representation, where $j = j_1 j_2 ... j_n$, we can eventually derive a product expression (Nielsen and Chuang 2010, p. 218) for the QFT using tensor operations:

$$|j\rangle = \frac{(|0\rangle + e^{2\pi i 0.j_n}|0\rangle + e^{2\pi i 0.j_{n-1}j_n}|1\rangle)...(|0\rangle + e^{2\pi i 0.j_1 j_2 ... j_n})}{2^{\frac{n}{2}}}$$

Where we use the notation $0.j_l j_{l+1} ... j_m = \frac{j_l}{2} + \frac{j_{l+1}}{4} + ... + \frac{j_m}{2^{m-l+1}}$ which represents a binary fraction.
To see what this visually looks like for n = 3 (a 3-qubit QFT), we present the following circuit:



Note that the initial state is represented as $|q_0 q_1 q_2\rangle$. This figure was created using Qiskit (Javadi-Abhari et al. 2024) and verified using NVIDIA's CUDA-Q v13.2 (NVIDIA 2026), applying a comparison against the QFT matrix that we apply on $|000\rangle$ to obtain our expected result (Qiskit and CUDA-Q are Python packages).

We can see the SWAP gate being used at the end of the QFT, where at most $\frac{n}{2}$ SWAP gates are required for a given QFT applied on n qubits (Nielsen and Chuang 2010, p. 220).

We can compare the QFT (on n qubits) to the classical algorithm, the Fast Fourier Transform (FFT) (on N samples) (Nielsen and Chuang 2010, p. 220):

$$[QFT, \quad \Theta(n^2)] \qquad [FFT, \quad \Theta(N \log_2 N)], \qquad N = 2^n$$

From this we can see that the QFT is much faster, while achieving fundamentally the same outcome.

If I had more time, I would consider the following:

- Shor's Algorithm, $O((\log(N))^3)$, for factorising integers into their primes.
- Grover's Algorithm, $O(\sqrt{N})$, for finding the shortest route in a network of N possible routes.
- Limitations of the QFT (hardware, availability)

As per college guidelines, I acknowledge that I have used GenAI for the purpose of searching for literature (textbooks, Python packages), solving LaTeX issues (formatting, packages) and getting feedback on my work (including GitHub files).

Citations, Derivations and Python